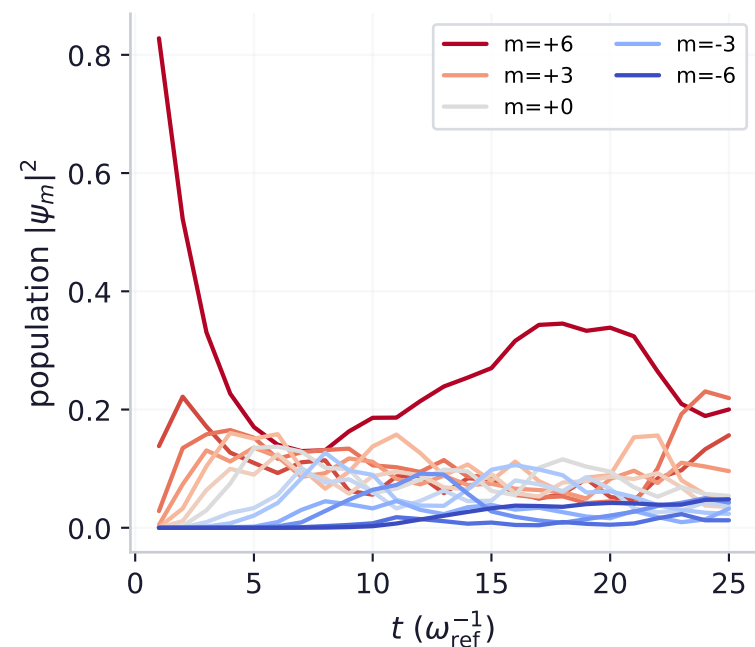
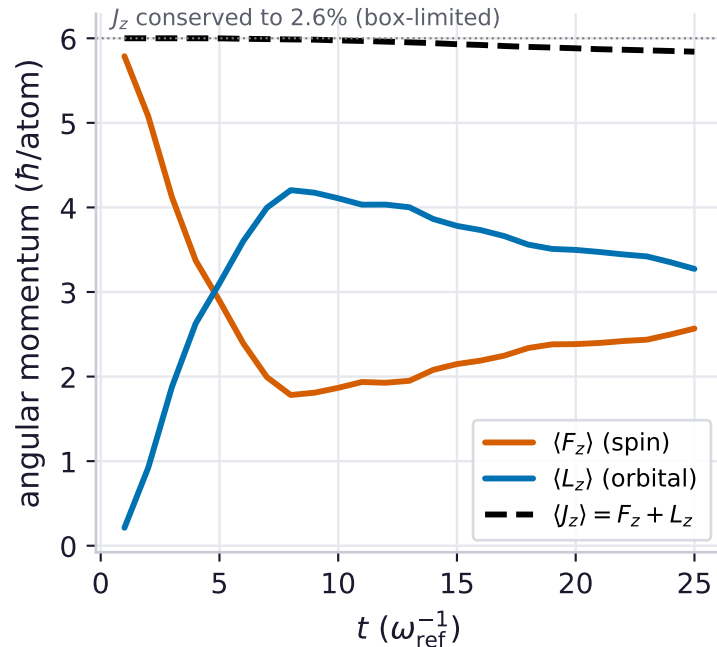


Einstein-de Haas in ^{151}Eu ($F=6$) dipolar BEC: field quench drives spin $\rightarrow$ orbital transfer (real vortices, no rotation yet)

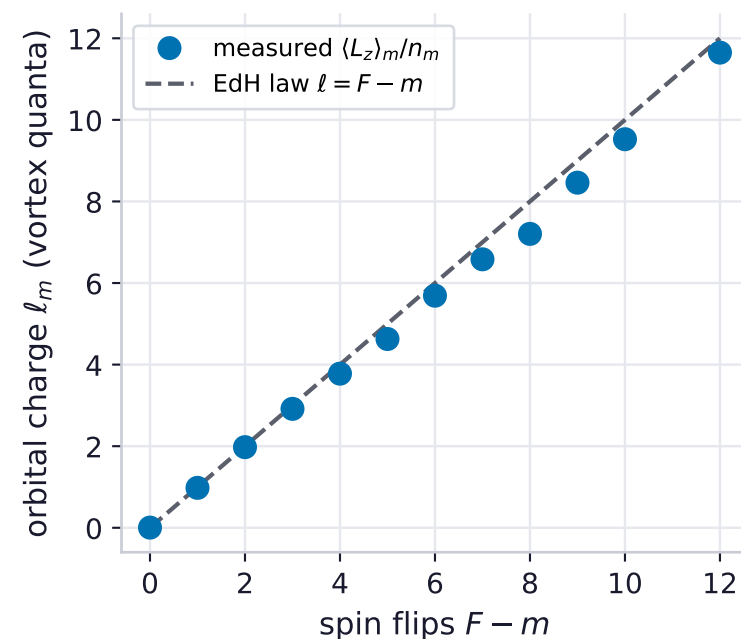
(a) spin excitation: m-ladder cascade



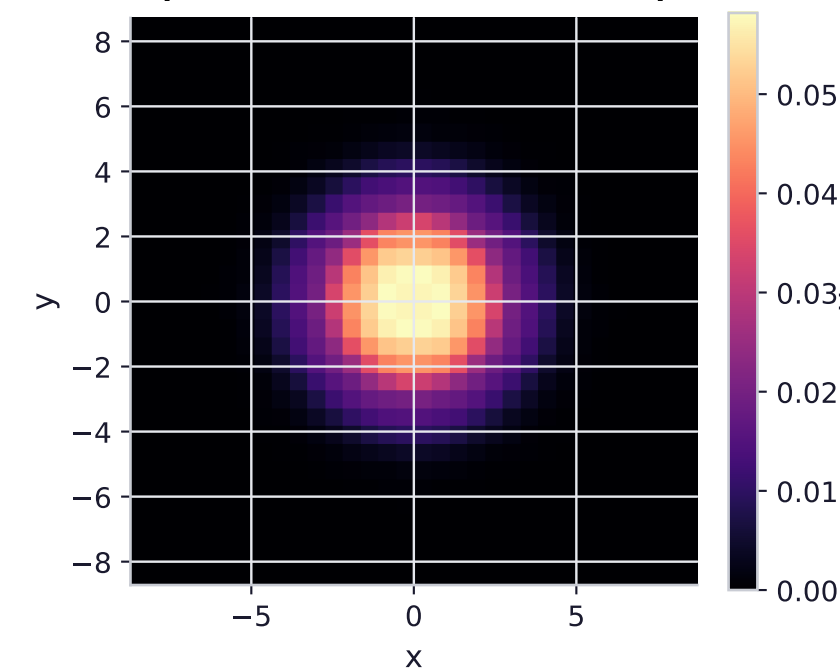
(b) Einstein-de Haas: spin $\rightarrow$ orbital, J_z conserved



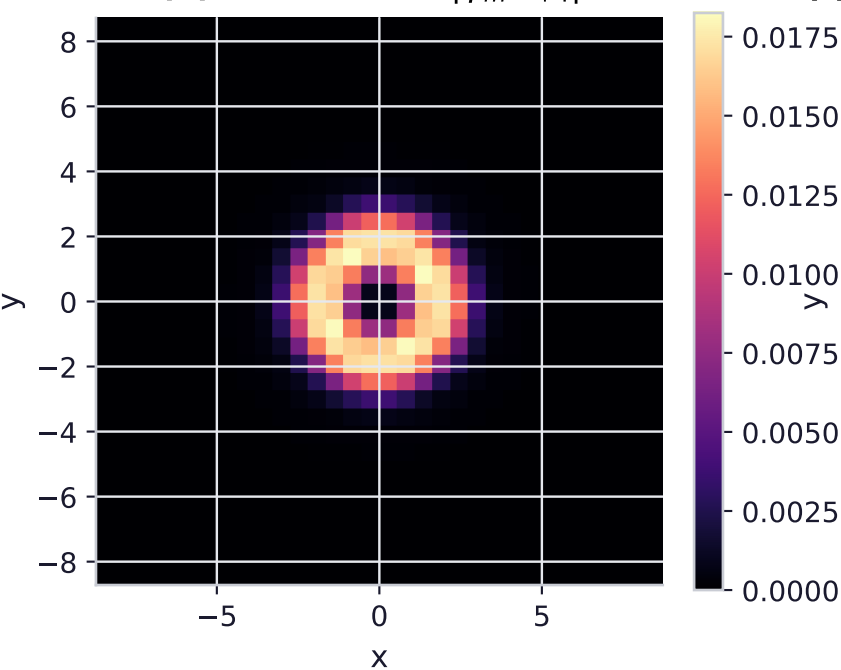
(c) quantised vortices: $\ell_m = F - m$



**(d) total column density, $t=25$
(vortex cores = dark holes)**



(e) transferred $|\psi_{m=+4}|^2$



(f) phase of $\psi_{m=+4}$ (2π winding = vortex)

